

Nash's theorem via Günther's trick:
three lectures

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These are polished version of my lecture notes. The reader is assumed to be familiar with smooth manifolds (charts, atlas, coverings, partitions of unity, and tensor fields); we also use Schauder estimates for solutions of elliptic partial differential equations, but they could be used as a black box. I want to thank Michael Gromov, Fedor Nazarov, and Deane Yang for their help.

Contents

1 Formulation and approximation	5
A. Induced metric 5 ; B. Q-form 7 ; C. Disc covering 7 ; D. Nash's twist 8 ; E. Approximate version 9 ; F. A pseudoeuclidean degression 11 .	
2 Reductions	13
A. Restricted theorem 13 ; B. General metric on torus 14 ; C. From immersion to embedding 14 ; D. From torus to compact 14 ; E. From compact to noncompact 15 .	
3 The proofs	17
A. Free maps 17 ; B. Formulations 18 ; C. Perturbation 19 ; D. Proof of Nash's lemma 21 ; E. Proof of Günther's lemma 22 .	
Semisolutions	25
Bibliography	29

Lecture 1

Formulation and approximation

Riemannian geometry has two fundamental theorems. One is about the existence and uniqueness of Levi-Civita connection and the other refers to the result of John Nash [7] that *any Riemannian manifold is isometric to a smooth submanifold in a Euclidean space*. In other words, Riemannian manifolds can be defined as a submanifolds of Euclidean space.

We will present a simplified proof discovered by Matthias Günther [1] with minor improvements due to Deane Yang, Terence Tao, and Ralph Howard [3, 8, 11]. Günther's approach was also used in books by Michael Taylor [9] and by Qing Han and Jia-Xing Hong [2].

A Induced metric

Given a smooth n -dimensional manifold Ω , we will denote by $T\Omega$ and $T_p\Omega$ its tangent bundle and tangent space at a point $p \in \Omega$, respectively.

Recall that a metric tensor, say g , on Ω is a smooth field of bilinear symmetric forms on $T\Omega$. Here and further, *smooth* means C^∞ .

Let $\bar{\Omega}$ be another smooth manifold and let $w: \Omega \rightarrow \bar{\Omega}$ be a smooth map. Recall that derivative of w in direction of a tangent vector $x \in T_p\Omega$ is denoted by xw and

$$xw = dw(x);$$

that is, the derivative of w in the direction x is the differential $dw: T\Omega \rightarrow T\bar{\Omega}$ of w evaluated at x . Note that $xw \in T_{w(p)}\bar{\Omega}$.

Suppose that $\bar{\Omega}$ is equipped with a metric tensor $\bar{g}$. The metric tensor g on Ω defined by

$$g(X, Y) = \bar{g}(Xw, Yw)$$

will be called induced metric tensor, or pullback of $\bar{g}$ to Ω . This relation can be also written as $g = w^*\bar{g}$. Equivalently, we can say that w is an isometric map from (Ω, g) to $(\bar{\Omega}, \bar{g})$.

A metric tensor g is called Riemannian if it is *positive*; that is, $g(X, X) > 0$ for any tangent vector $X \neq 0$. A smooth manifold equipped with a Riemannian metric is called Riemannian manifold.

Note that if $g = w^*\bar{g}$ is Riemannian metric, then w has to be regular; that is, the differential $d_p w$ has rank n at any point $p \in \Omega$. In particular, w is an immersion.

The space $\mathbb{R}^d$ will be considered with metric tensor defined by the standard scalar product defined by

$$\bar{g}(X, Y) := \langle x, y \rangle = x_1 \cdot y_1 + \cdots + x_d \cdot y_d,$$

where x_i and y_i denote the coordinates of vectors $x, y \in \mathbb{R}^d$. Note that $\bar{g}$ is Riemannian.

1.1. Main theorem. *Any n -dimensional Riemannian manifold (Ω, g) admits an isometric embedding into a Euclidean space $\mathbb{R}^d$ for some d . Moreover, d can be bounded in terms of n .*

We will not make any effort to optimize the bound on d in terms of n . Tracing the estimates in our argument yields $d \leq 10 \cdot n^3$. The lower bound from the following exercise might be optimal, the known upper bounds are not that far today.

1.2. Exercise. *Show that d in the main theorem has to be at least $n \cdot (n + 1)/2$.*

1.3. Exercise. *Construct a metric tensor g on the torus $\mathbb{T}^2 = \mathbb{S}^1 \times \mathbb{S}^1$ such that $g(X, X) \geq 0$ for any tangent vector X , but there is no smooth map $w: \mathbb{T}^2 \rightarrow \mathbb{R}^{10}$ with the induced metric g .*

1.4. Exercise. *Let w be a smooth map from a smooth Riemannian manifold (Ω, g) to the Euclidean space $\mathbb{R}^d$. Show that w is isometric if and only if it is length-preserving; that is, if*

$$\text{length } \gamma = \text{length}(w \circ \gamma)$$

for any curve γ in Ω .

B Q-form

Recall that Ω is a smooth n -dimensional manifold.

Let us define a symmetric bilinear form Q that swallows two C^1 -smooth maps $v, w: \Omega \rightarrow \mathbb{R}^d$ and spits the following metric tensor on Ω

$$g(X, Y) \equiv \frac{1}{2} \cdot [\langle Xv, Yw \rangle + \langle Xw, Yv \rangle].$$

Note that if the maps v and w are C^k -smooth, then $g = Q(v, w)$ is C^{k-1} -smooth.

1.5. Observation. *A metric tensor g is induced by a smooth map $w: \Omega \rightarrow \mathbb{R}^d$ if and only if $g = Q(w, w)$.*

The components of $g = Q(w, w)$ in a local chart can be written as

$$g_{ij} = Q_{ij}(w, w) = \langle \partial_i w, \partial_j w \rangle.$$

Let Ω be a smooth manifold. Given two smooth maps $v_1: \Omega \rightarrow \mathbb{R}^{d_1}$ and $v_2: \Omega \rightarrow \mathbb{R}^{d_2}$, denote by $w = v_1 \oplus v_2$ the map

$$w: \Omega \rightarrow \mathbb{R}^{d_1+d_2} = \mathbb{R}^{d_1} \oplus \mathbb{R}^{d_2}$$

defined by $w: p \mapsto (v_1(p), v_2(p))$.

1.6. Exercise. *Show that*

$$Q(v_1 \oplus v_2, v_1 \oplus v_2) = Q(v_1, v_1) + Q(v_2, v_2)$$

for any two smooth maps $v_1: \Omega \rightarrow \mathbb{R}^{d_1}$ and $v_2: \Omega \rightarrow \mathbb{R}^{d_2}$.

Recall that support of a map is the closure of the set of its arguments with nonzero values.

1.7. Exercise. *Let $v_1, v_2: \Omega \rightarrow \mathbb{R}^d$ be smooth maps defined on a smooth manifold Ω . Suppose v_1 and v_2 have disjoint supports. Show that*

$$Q(v_1 + v_2, v_1 + v_2) = Q(v_1, v_1) + Q(v_2, v_2).$$

C Disc covering

Let Ω be a smooth n -dimensional manifold. A subset D in Ω will be called smooth n -disc if there is a neighborhood N of D and a smooth embedding $N \rightarrow \mathbb{R}^n$ that sends D to the unit ball.

1.8. Claim. *Let $\{U_\alpha\}$ be an open cover of a smooth n -dimensional manifold Ω . Then there is a countable collection of smooth n -discs $\{D_i\}$ in Ω such that the following condition hold.*

- (a) The interiors D_i° of D_i cover the whole Ω .
- (b) For each D_i there is α such that $D_i \subset U_\alpha$.
- (c) The collection $\{D_i\}$ can be colored into $n + 1$ colors such that distinct n -discs of the same color are disjoint. More presicely, the index set $\mathfrak{I}$ of $\{D_i\}$ can be subdivided into $n + 1$ subsets $\mathfrak{I}_0, \dots, \mathfrak{I}_n$ such that for any $\mathfrak{I}_i$ and any distict $i, j \in \mathfrak{I}_i$ we have $D_i \cap D_j = \emptyset$.

Proof. Recall that a star of a vertex i in a simplicial complex is a union of all open simplices that have i as a vertex.

Choose a triangulation τ of Ω such that each star of τ lies in some of U_α . (To prove the existance, we may use that any smooth manifold admits a triangulation and then apply [10, Theorem 35].) Let τ' be the baricetric subdivision of τ .

Choose a vertex i of τ' . Note that the star of i admits a smooth parametrization by an open n -disc. Passing to a slightly smaller disc in this parametrization produce a smooth n -disc D_i . This procedures can be arranged in such a way that the inerioris D_i° for all vertexes i of τ' cover Ω . Since D_i lies in the star of i , we have condition (b).

Recall that each vertex of τ' corresponds to a simplex in τ . Let us color the vertices of τ' into $n + 1$ colors $0, \dots, n$ so that vertex of color k correspond to a k -simplex in τ . Observe that the induced coloration of the n -discs satisfies the condition (c). $\square$

D Nash's twist

Let φ and ψ be smooth functions on a smooth manifold Ω . Given $r > 0$, denote by $\mathbb{S}_r^1$ the circle of radius r in $\mathbb{R}^2$. The Nash's twist $\Theta: \Omega \rightarrow \mathbb{R}^2$ of the triple (r, φ, ψ) is defined as the composition

$$\Omega \xrightarrow{\psi} \mathbb{R} \xrightarrow{\ell_r} \mathbb{S}_r^1 \xrightarrow{\times \varphi} \mathbb{R}^2,$$

where $\ell_r: \mathbb{R} \rightarrow \mathbb{S}_r^1$ is a length-preserving covering map, say

$$\ell_r(x) = (r \cdot \cos \frac{x}{r}, r \cdot \sin \frac{x}{r}),$$

and $\times \varphi$ denotes the multiplication by φ ; so

$$\Theta(x) := \varphi(x) \cdot (\ell_r \circ \psi(x)).$$

Let us denote by $(d\psi)^2$ the metric tensor induced by $\psi: \Omega \rightarrow \mathbb{R}$; that is,

$$(d\psi)^2(X, Y) := d\psi(X) \cdot d\psi(Y) = (X\psi) \cdot (Y\psi).$$

1.9. Claim. *The Nash's twist Θ for triple (r, φ, ψ) induces the metric*

$$g = \varphi^2 \cdot (d\psi)^2 + r^2 \cdot (d\varphi)^2.$$

Computations. Suppose that x is a tangent vector on Ω , then

$$\begin{aligned} x\Theta &= x(\varphi \cdot \ell_r \circ \psi) = \\ &= (x\varphi) \cdot (\ell_r \circ \psi) + \varphi \cdot (\ell'_r \circ \psi) \cdot (x\psi). \end{aligned}$$

Observe that $|\ell_r| = r$, $|\ell'_r| = 1$, and $\ell_r \perp \ell'_r$. Therefore,

$$g(x, y) = \langle x\Theta, y\Theta \rangle = r^2 \cdot (x\varphi) \cdot (y\varphi) + \varphi^2 \cdot (x\psi) \cdot (y\psi). \quad \square$$

E Approximate version

The following statement can be regarded as an approximate version of the main theorem (1.1).

1.10. Proposition. *Let g be a Riemannian metric on a smooth n -manifold Ω . Then there is a metric tensor h on Ω and one-parameter family of smooth maps $w_t: \Omega \rightarrow \mathbb{R}^d$ such that*

$$Q(w_t, w_t) = g + t \cdot h$$

for any $t > 0$.

Note that $Q(w_t, w_t)$ converges to g as $t \rightarrow 0$. Therefore, it is tempting to pass to the limit of w_t as $t \rightarrow 0$. However, as you will see, the maps w_t constructed in the proof converge to a constant; so the limit map does not solve 1.1.

Let $w: \Omega \rightarrow \mathbb{R}^d$ be a smooth map. Suppose that $w_1, \dots, w_d$ are coordinate functions of w . Then the induced metric tensor g can be written as

$$\textbf{1} \quad g = Q(v, w) = (dw_1)^2 + \dots + (dw_d)^2.$$

In order to prove the main theorem, we need to find functions $w_1, \dots, w_d: \Omega \rightarrow \mathbb{R}$ that solves **1** for a given Riemannian metric. The weaker form **2** of **1** will play a key role in the proof of 1.10.

1.11. Proposition. *Let (Ω, g) be a compact n -dimensional Riemannian manifold. Then there are smooth functions $\varphi_1, \dots, \varphi_d, \psi_1, \dots, \psi_d$ on Ω such that*

$$\textbf{2} \quad g = \varphi_1^2 \cdot (d\psi_1)^2 + \dots + \varphi_d^2 \cdot (d\psi_d)^2.$$

Moreover, we can assume that $d \leq 2 \cdot (n+1) \cdot n^2$.

Proof. The metric tensor g can be written in local cart $(x_1, \dots, x_n)$ as

$$\textcircled{3} \quad g = \sum_{i,j} g_{ij} \cdot dx_i \cdot dx_j,$$

where $g_{ij} = g_{ji}$ are smooth functions of $(x_1, \dots, x_n)$.

Since g is Riemannian, at a given point $p \in \Omega$, we can choose a chart so that ∂_i are orthonormal at p ; in other words, $g_{ii} = 1$ and $g_{ij} = 0$ at p for all $i \neq j$. Since g_{ij} are smooth, given $\varepsilon > 0$, we can choose a neighborhood $U \ni p$ such that

$$\textcircled{4} \quad g_{ii} \leq 1 \pm \varepsilon \quad \text{and} \quad g_{ij} \leq \pm \varepsilon$$

for all $i \neq j$ and any point in U .

Observe that

$$\pm dx_i \cdot dx_j = \frac{1}{2} \cdot (dx_i \pm dx_j)^2 - \frac{1}{2} \cdot (dx_i)^2 - \frac{1}{2} \cdot (dx_j)^2.$$

Let us plug this formula in $\textcircled{3}$ and take $\textcircled{4}$ into account. Assuming that ε is small, we will get that g is a linear combination with positive coefficients of the metric tensors $(dx_i)^2$ and $(dx_i \pm dx_j)^2$ for all i and j . In other words, we can take the functions x_i and $x_i \pm x_j$ for all $i < j$ as our functions $\psi_1, \dots, \psi_d$ (these should be extended smoothly to the whole manifold) and find $\varphi_1, \dots, \varphi_d$ such that $\textcircled{2}$ holds in a neighborhood of p .

Applying a partition-of-unity argument, we get the global statement.

Note that it is sufficient to take n^2 functions in each chart; so, if Ω is covered by m charts, then one can take $d = n^2 \cdot m$ functions. Moreover, the same holds if the charts can be colored in m colors such that distinct chars of the same color do not intersect. Therefore, Claim 1.8 implies that $d = n^2 \cdot (n+1)$ is sufficient. $\square$

Proof of 1.10. Suppose that $\varphi_1, \dots, \varphi_d, \psi_1, \dots, \psi_d$ are provided by 1.11. Let $r = \sqrt{t}$; denote by Θ_i the Nash's twist for the triple (r, φ_i, ψ_i) . By 1.9,

$$w_t = \Theta_1 \oplus \dots \oplus \Theta_d$$

is the needed map with $h = (d\varphi_1)^2 + \dots + (d\varphi_d)^2$. $\square$

F A pseudoeuclidean degression

Let us denote by $\mathbb{R}^r \ominus \mathbb{R}^s$ the pseudoeuclidean space with signature (r, s) ; that is, the space $\mathbb{R}^{r+s}$ with the scalar product

$$\langle x, y \rangle = x_1 \cdot y_1 + \cdots + x_s \cdot y_s - x_{s+1} \cdot y_{s+1} - \cdots - x_{s+r} \cdot y_{s+r},$$

where x_i and y_i denote the coordinates of vectors x and y in $\mathbb{R}^{r+s}$.

1.12. Problem. *Show that any metric g on a smooth manifold Ω is induced by a smooth embedding $\Omega \rightarrow \mathbb{R}^r \ominus \mathbb{R}^s$ for some large r and s .*

Solution based on Nash's theorem. Note that $g = g_1 - g_2$ for two Riemannian metrics g_1 and g_2 on Ω . By Nash's theorem, we can find two smooth embeddings w_1 and w_2 of Ω to $\mathbb{R}^d$ with induced metrics g_1 and g_2 respectively. It remains to observe that g is induced by

$$w_1 \ominus w_2: p \mapsto (w_1(p), w_2(p)) \in \mathbb{R}^d \ominus \mathbb{R}^d.$$

□

The above solution uses the main theorem, which we did not prove. The following solution is more direct; it uses only Nash's twist.

Simpler solution. Assume g is Riemannian and $\varphi_1, \dots, \varphi_d, \psi_1, \dots, \psi_d$ are provided by 1.11. Let Θ_i be Nash's twist for the triple $(1, \varphi_i, \psi_i)$, and

$$\begin{aligned} h &= (d\varphi_1)^2 + \cdots + (d\varphi_d)^2, \\ \Theta &= \Theta_1 \oplus \cdots \oplus \Theta_d: \Omega \rightarrow \mathbb{R}^{2 \cdot d}, \\ \varphi &= \varphi_1 \oplus \cdots \oplus \varphi_d: \Omega \rightarrow \mathbb{R}^d. \end{aligned}$$

According to 1.9, Θ induces $g + h$. Note that φ induces h . Therefore $w = \Theta \ominus \varphi: \Omega \rightarrow \mathbb{R}^{2 \cdot d} \ominus \mathbb{R}^d$ induces g .

To solve the general case, present the metric as a difference of two Riemannian metrics $g = g_1 - g_2$. From above, there are a smooth embeddings w_1 and w_2 into pseudoeuclidean spaces that induces g_1 and g_2 respectively. It remains to observe that $w_1 \ominus w_2$ induces g . □

1.13. Exercise. *Let ψ be a smooth isometric immersion of a Riemannian manifold (Ω, g) into the sphere $\mathbb{S}^{d-1} \subset \mathbb{R}^d$ and let $\varphi: \Omega \rightarrow \mathbb{R}$ is a smooth function. Show that $\varphi^2 \cdot g$ is induced by $\varphi \cdot \psi \ominus \varphi$.*

Lecture 2

Reductions

In this lecture, we reduce the main theorem (1.1) to its restricted version (2.1), which is immersion theorem for very special metrics on tori. The reduction is made in the following steps:

- ◊ In Section 2B, we show that the restricted theorem (2.1) implies the immersion theorem for any metric on torus (2.2).
- ◊ In Section 2C, we show that immersion theorem for torus (2.2) implies the embedding theorem for torus (2.3).
- ◊ In Section 2D, we show that if main theorem holds for tori (2.3), then it holds for any compact manifolds (2.4).
- ◊ In Section 2E, we show that if the main theorem holds for compact manifolds (2.4), then it holds all manifolds.

A Restricted theorem

Let us denote by $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$ the n -dimensional torus.

The torus is a smooth manifold; it comes with a global periodic coordinates which can be used to define the partial derivatives $\partial_1, \dots, \partial_n$. A metric tensor on $\mathbb{T}^n$ can be written in these coordinates; so we can (and will) think that a metric tensor is a smooth map $g: \mathbb{T}^n \rightarrow \mathbb{R}^N$; here and further $N := \frac{n \cdot (n+1)}{2}$. These observations make it easier to work with torus, then with general manifold.

2.1. Restricted theorem. *There is a Riemannian metric g_0 on $\mathbb{T}^n$ such that for any metric tensor h on $\mathbb{T}^n$ the metric $g_0 + t \cdot h$ is induced by a smooth map $\mathbb{T}^n \rightarrow \mathbb{R}^d$ for any sufficiently small $t > 0$. Moreover, the dimension d can be bounded in terms of n .*

This theorem will be proved in the next lecture; see Section 3C.

B General metric on torus

2.2. Claim. *Given a Riemannian metric g on $\mathbb{T}^n$ there is an isometric immersion $(\mathbb{T}^n, g) \looparrowright \mathbb{R}^d$ for some d that depends only on n .*

Proof. Let g_0 be the metric from 2.1. Applying rescaling if necessary, we can assume that $g_0 < g$; that is, $g_0(X, X) < g(X, X)$ for any tangent vector $X \neq 0$. In particular, $g_1 = g - g_0$ is a Riemannian metric.

By 1.10, we have a metric h such that for any $t > 0$ there is a smooth map $w_1: \mathbb{T}^n \rightarrow \mathbb{R}^{d_1}$ that induces $g_1 + t \cdot h$.

By 2.1, for any small $t > 0$ there is a smooth map $w_0: \mathbb{T}^n \rightarrow \mathbb{R}^{d_0}$ such that induce metric $g_0 - t \cdot h$.

By 1.6, the metric tensor

$$g = g_0 - t \cdot h + g_1 + t \cdot h$$

is induced by the map $w_0 \oplus w_1$. □

C From immersion to embedding

2.3. Claim. *Given a Riemannian metric g on $\mathbb{T}^n$, there is an isometric embedding $(\mathbb{T}^n, g) \hookrightarrow \mathbb{R}^d$ for some d that depends only on n .*

Proof. Choose your favorite smooth embedding $w_1: \mathbb{T}^n \rightarrow \mathbb{R}^{n+1}$. Let $g_1 = Q(w_1, w_1)$; in other words, g_1 is the metric induced by w_1 . Since w_1 is an embedding, g_1 must be Riemannian.

We can assume that $g > g_1$; in other words the metric $g_2 = g - g_1$ is Riemannian. If this is not the case, pass to a rescaling $\varepsilon \cdot w_1$ of w_1 for small $\varepsilon > 0$.

By 2.2, we have a smooth isometric immersion $w_2: (\mathbb{T}^n, g_2) \rightarrow \mathbb{R}^d$. Note that

$$w = w_1 \oplus w_2: (\mathbb{T}^n, g = g_1 + g_2) \rightarrow \mathbb{R}^{d+2 \cdot n+1}$$

is an isometric embedding. Indeed, w is isometric by 1.6, and it is an embedding since w_1 is. □

D From torus to compact

2.4. Claim. *Any compact n -dimensional Riemannian manifold (Ω, g) admits an isometric embedding into $\mathbb{R}^d$ for some d that depends only on n .*

This part of reduction relies on the following exercise.

2.5. Exercise. *Let Ω be a closed submanifold in the torus $\mathbb{T}^n$. Show that any Riemannian metric on Ω is a restriction of a Riemannian metric on $\mathbb{T}^n$.*

Reduction 2.4 to 2.3. Since Ω is compact, it can be smoothly embedded in $\mathbb{T}^{2 \cdot n+1}$.

Indeed, by the Whitney embedding theorem, there is a smooth embedding $\iota: \Omega \rightarrow \mathbb{R}^{2 \cdot n+1}$. Let g_1 be the metric on Ω induced by ι . Since Ω is compact, we can assume in addition that $g > g_1$ (that is, $g - g_1$ is Riemannian) and its image lies in a small ball; if not, then pass to the rescaling $\varepsilon \cdot \iota$ of ι for small $\varepsilon > 0$. Composing ι with the covering map $\mathbb{R}^{2 \cdot n+1} \rightarrow \mathbb{T}^{2 \cdot n+1}$, we get the required embedding.

Now, we can assume that Ω is a compact submanifold of $\mathbb{T}^{2 \cdot n+1}$. By 2.5, any Riemannian metric g on Ω is the restriction of a Riemannian metric, say $\bar{g}$, on $\mathbb{T}^{2 \cdot n+1}$.

Choose an isometric smooth embedding $w: (\mathbb{T}^{2 \cdot n+1}, \bar{g}) \rightarrow \mathbb{R}^d$ provided by 2.3. Observe that the composition $f \circ \iota: (\Omega, g) \rightarrow \mathbb{R}^d$ is isometric. $\square$

E From compact to noncompact

This part is known as Nash's reduction; it appears in [7, Part D] Let us start with two exercises.

2.6. Exercise. *Let Ω be a connected noncompact manifold. Show that it admits a smooth proper bounded embedding in $\mathbb{R}^d \setminus \{0\}$. Moreover, given a Riemannian metric g on Ω , we can assume that the induced metric of this embedding is smaller than g .*

2.7. Exercise. *Suppose that $\{D_i\}$ is a collection of smooth n -discs on a smooth n -dimensional manifold Ω provided by 1.8. Show that for any Riemannian metric g on Ω there is a collection of Riemannian metrics h_i on $\mathbb{S}^n$ and a smooth maps $r_h: \Omega \rightarrow \mathbb{S}^n$ such that*

- (a) *The restriction $r_i|_{D_i^\circ}$ is a smooth embedding,*
- (b) *r_i maps $\Omega \setminus D_i^\circ$ to the south pole of $\mathbb{S}^n$*
- (c) *$g = \sum_i r_i^* h_i$.*

Reduction of 1.1 to 2.4. Let $w: \Omega \rightarrow \mathbb{R}^d$ be an embedding provided by 2.6 for metric $\frac{1}{2} \cdot g$. Let g_0 be the metric induced by w ; note that $g_1 = g - g_0$ is Riemannian.

Let us apply 2.7 to (Ω, g_1) ; denote by r_i , h_i , and D_i the obtained maps, metrics and smooth n -discs. Since $\mathbb{S}^n$ is compact, there is an isometric embedding $w_i: (\mathbb{S}^n, h_i) \rightarrow \mathbb{R}^d$; we can assume that $w_i \circ r_i$ maps all points in $\Omega \setminus D_i^\circ$ to the origin on $\mathbb{R}^d$; in particular, each composition $w_i \circ r_i$ has compact support in D_i° .

Let $\mathcal{I}_0, \dots, \mathcal{I}_n$ be the coloring of the index set provided by 1.8. Set

$$s_i = \sum_{i \in \mathcal{I}_i} w_i \circ r_i \quad \text{and} \quad s = s_1 \oplus \dots \oplus s_{n+1}.$$

By 1.7 and 1.6,

$$\begin{aligned} Q(w \oplus s, w \oplus s) &= Q(w, w) + \sum_i Q(s_i, s_i) = \\ &= g_0 + \sum_i Q(w_i \circ r_i, w_i \circ r_i) = \\ &= g_0 + \sum_i r_i^* h_j \\ &= g_0 + g_1 = g. \end{aligned}$$

Since w is an emebdding, so is $w \oplus s$, hence the result. □

Lecture 3

The proofs

In this lecture we formulate and prove three lemmas that imply the restricted version of the main theorem 2.1 and therefore the main theorem 1.1.

A Free maps

As before, Ω is a smooth n -dimensional manifold.

Recall that a C^1 -map $f: \Omega \rightarrow \mathbb{R}^d$ is called regular if its differential df has rank n at each point. In other words, for any point $p \in \Omega$ and some (and therefore any) local coordinates at p all first-order partial derivatives $\partial_1 f, \dots, \partial_n f$ are linearly independent.

A C^2 -map $f: \Omega \rightarrow \mathbb{R}^d$ is called free (or two-regular) if an analogous property holds for first *and second* partial derivatives; that is, for any point $p \in \Omega$ and some (and therefore any) local coordinates at p all $n + \frac{n \cdot (n+1)}{2}$ partial derivatives $\partial_i f, \partial_{ij} f$ for $i \leq j$ are linearly independent.

3.1. Exercise. *Show that freeness of map $f: \Omega \rightarrow \mathbb{R}^d$ is independent on choice of charts on Ω .*

3.2. Exercise. *Show that there is no free map $\mathbb{S}^2 \rightarrow \mathbb{R}^4$.*

3.3. Exercise. *Let $f: \Omega \rightarrow \mathbb{R}^{d_1}$ and $w: \Omega \rightarrow \mathbb{R}^{d_2}$ be smooth maps defined on a smooth manifold Ω . Show that if f is free, then so is $f \oplus w$.*

Let ψ be a smooth function on a manifold Ω . Denote by Θ_ψ the smooth map $\Omega \rightarrow \mathbb{R}^2$, defined by

$$\Theta_\psi(x) = (\cos[\psi(x)], \sin[\psi(x)]).$$

In other words, Θ_ψ is the composition of two maps

$$\Omega \xrightarrow{\psi} \mathbb{R} \xrightarrow{\ell} \mathbb{S}^1 \subset \mathbb{R}^2,$$

where $\ell: \mathbb{R} \rightarrow \mathbb{S}^1$ is the natural covering maps $y \mapsto (\cos y, \sin y)$. Note that Θ_ψ is the Nash's twist for the triple $(1, 1, \psi)$.

3.4. Exercise. Consider the (x, y) -plane $\mathbb{R}^2$. Show that the map

$$\Theta_x \oplus \Theta_y \oplus \Theta_{x+y}: \mathbb{R}^2 \rightarrow \mathbb{R}^6 = \mathbb{R}^2 \oplus \mathbb{R}^2 \oplus \mathbb{R}^2$$

is free.

Generalize the statement to maps $\mathbb{R}^n \rightarrow \mathbb{R}^{n \cdot (n+1)}$.

3.5. Exercise. Let $w: \Omega \rightarrow \mathbb{R}^s$ be a regular smooth map and $f: \mathbb{R}^s \rightarrow \mathbb{R}^d$ is free. Show that the composition $f \circ w: \Omega \rightarrow \mathbb{R}^d$ is free.

Apply the Whitney embedding theorem and 3.4 to conclude that any smooth manifold admits a free embedding into a Euclidean space.

B Formulations

Recall that our torus $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$ is equipped with of periodic coordinates and globally defined partial derivatives $\partial_1, \dots, \partial_n$; furthermore, a metric tensor on $\mathbb{T}^n$ can be described by a smooth map $g: \mathbb{T}^n \rightarrow \mathbb{R}^N$, where $N := \frac{n \cdot (n+1)}{2}$.

Fix $\alpha \in (0, 1)$, say $\alpha = \frac{1}{2}$. Denote by $|x - y|_{\mathbb{T}^n}$ the standard distance between points $x, y \in \mathbb{T}^n$. Let us recall the definition of Hölder norms of a function $u: \mathbb{T}^n \rightarrow \mathbb{R}$

$$\begin{aligned} \|u\|_{0,\alpha} &:= \sup_{x \in \mathbb{T}^n} \{ |u(x)| \} + \sup_{x, y \in \mathbb{T}^n} \left\{ \frac{|u(x) - u(y)|}{|x - y|_{\mathbb{T}^n}^\alpha} \right\}, \\ \|u\|_{k,\alpha} &:= \max_{|\mathcal{I}| \leq k} \{ \|\partial_{\mathcal{I}} u\|_{0,\alpha} \}, \end{aligned}$$

where $\mathcal{I} = (i_1, \dots, i_n)$ denotes multi-index; so, $\partial_{\mathcal{I}} := \partial_1^{i_1} \dots \partial_n^{i_n}$ and $|\mathcal{I}| := i_1 + \dots + i_n$. The space of functions with finite norm $\|\cdot\|_{k,\alpha}$ is called the (k, α) -Hölder space; it will be denoted by $C^{k,\alpha}(\mathbb{T}^n, \mathbb{R})$.

3.6. Exercise. Let $v \in C^\infty(\mathbb{T}^n, \mathbb{R})$. Show that there is a constant c such that

$$\|u \cdot v\|_{k,\alpha} \leq c \cdot \|u\|_{k,\alpha}$$

for any $u \in C^{k,\alpha}(\mathbb{T}^n, \mathbb{R})$.

Let us define $C^{k,\alpha}(\mathbb{T}^n, \mathbb{R}^d)$ as the space of functions $u: \mathbb{T}^n \rightarrow \mathbb{R}^d$ with all coordinate functions $u_1, \dots, u_d$ from $C^{k,\alpha}(\mathbb{T}^n, \mathbb{R})$, and set

$$\|u\|_{k,\alpha} := \max_i \{ \|u_i\|_{k,\alpha}, \dots, \|u_d\|_{k,\alpha} \}.$$

From now on, fix a smooth free map $f: \mathbb{T}^n \rightarrow \mathbb{R}^d$; it exists by 3.5. Furthermore, let

$$L(w) := 2 \cdot Q(f, w).$$

3.7. Nash's lemma. *The linear operator*

$$L: C^\infty(\mathbb{T}^n, \mathbb{R}^d) \rightarrow C^\infty(\mathbb{T}^n, \mathbb{R}^N)$$

has a right inverse

$$M: C^\infty(\mathbb{T}^n, \mathbb{R}^N) \rightarrow C^\infty(\mathbb{T}^n, \mathbb{R}^d)$$

such that

$$\|Mh\|_{2,\alpha} \leq a \cdot \|h\|_{2,\alpha}$$

for some constant a .

3.8. Günther's lemma. *There is a symmetric bilinear form*

$$\tilde{Q}: C^\infty(\mathbb{T}^n, \mathbb{R}^d) \times C^\infty(\mathbb{T}^n, \mathbb{R}^d) \rightarrow C^\infty(\mathbb{T}^n, \mathbb{R}^d)$$

such that

$$L\tilde{Q} = Q,$$

$$\|\tilde{Q}(w, w)\|_{2,\alpha} \leq b_2 \cdot \|w\|_{2,\alpha}^2,$$

$$\|\tilde{Q}(w, w)\|_{k,\alpha} \leq b_2 \cdot \|w\|_{2,\alpha} \cdot \|w\|_{k,\alpha} + b_k \cdot \|w\|_{k-1,\alpha}^2$$

for any integer $k \geq 3$ and some fixed constants $b_2, b_3, \dots$

C Perturbation

Now we will finish the proof of Nash's theorem modulo the two lemmas formulated in the previous section. Recall that $f: \mathbb{T}^n \rightarrow \mathbb{R}^d$ is a fixed smooth free map and $L(w) := 2 \cdot Q(f, w)$.

3.9. Perturbation lemma. *There is $r > 0$ such that the equation*

$$\textcircled{1} \quad L(w) + Q(w, w) = h$$

has a smooth solution $w: \mathbb{T}^n \rightarrow \mathbb{R}^d$ for any smooth metric h such that $\|h\|_{2,\alpha} < r$. Moreover, we can assume that

$$\|w\|_{2,\alpha} \leq c \cdot \|h\|_{2,\alpha},$$

for some constant c .

Let g be the metric tensor g induced by f . Note that the equation

$$Q(f + w, f + w) = g + h$$

is equivalent to **1**. Indeed,

$$\begin{aligned} Q(f + w, f + w) &= Q(f, f) + 2 \cdot Q(f, w) + Q(w, w) \\ &= g + L(w) + Q(w, w). \end{aligned}$$

Therefore, the perturbation lemma implies the restricted theorem (2.1), which in turn implies the main theorem (1.1).

Proof of 3.9. Recall that M , $\tilde{Q}$, a , and b_2 are provided by 3.7 and 3.8. Set $R = \frac{1}{10 \cdot b_2}$ and $r = \frac{R}{10 \cdot a}$. Assume $\|h\|_{2,\alpha} \leq r$. By 3.8,

$$\Phi: w \mapsto Mh - \tilde{Q}(w, w)$$

is a contraction map from the closed ball $\bar{B}[0, R] \subset C^{2,\alpha}(\mathbb{T}^n, \mathbb{R}^d)$ to itself.

Consider the sequence of maps $w_0, w_1, \dots: \mathbb{T}^n \rightarrow \mathbb{R}^d$ such that $w_0 = 0$ and $w_{n+1} = \Phi(w_n)$ for any n . Note that w_n is smooth for each n . Since Φ is a contraction, w_n converges in $C^{2,\alpha}(\mathbb{T}^n, \mathbb{R}^d)$ as $n \rightarrow \infty$; denote its limit by w . Note that

$$\|w\|_{2,\alpha} \leq R \quad \text{and} \quad w = Mh - \tilde{Q}(w, w).$$

Since $LM = \text{id}$ and $L\tilde{Q} = Q$, applying L to both sides of the last equation, we get that w solves **1**.

To finish the proof it is sufficient to show the following.

2 *The sequence $\|w_n\|_{k,\alpha}$ is bounded for every integer $k \geq 2$.*

Indeed, since

$$\|w\|_{k,\alpha} \leq \lim_{n \rightarrow \infty} \|w_n\|_{k,\alpha}$$

it follows that $w \in C^{k,\alpha}$ for any k ; hence $w \in C^\infty$.

Claim **2** will be proved by induction on k . The base case $k = 2$ is proved already.

By the induction hypothesis, $\|w_n\|_{k-1,\alpha}$ is bounded. In particular, there is a constant c such that

$$\|Mh\|_{k-1,\alpha} + b_k \cdot \|w_n\|_{k-1,\alpha}^2 \leq c$$

for any n ; here b_k is the constant from 3.8.

By the second estimate in 3.8, we get

$$\begin{aligned} \|w_{n+1}\|_{k,\alpha} &\leq \|Mh\| + b_2 \cdot \|w_n\|_{2,\alpha} \cdot \|w_n\|_{k,\alpha} + b_k \cdot \|w_n\|_{k-1,\alpha}^2 \leq \\ &\leq c + b_2 \cdot R \cdot \|w_n\|_{k,\alpha} \leq \\ &\leq c + \frac{1}{10} \cdot \|w_n\|_{k,\alpha}. \end{aligned}$$

In particular,

$$\|w_n\|_{k,\alpha} \leq 2 \cdot c \quad \implies \quad \|w_{n+1}\|_{k,\alpha} \leq 2 \cdot c$$

for any n . Since $w_0 = 0$, Claim 2 holds for k . $\square$

D Proof of Nash's lemma

Recall that $f: \mathbb{T}^n \rightarrow \mathbb{R}^d$ is a fixed free map; in particular, f is regular and smooth.

The tangent space T_p is the n -dimensional space spanned by the first-order partial derivatives $\partial_1 f, \dots, \partial_n f$ at p . Denote by $T_p^\perp$ the normal complement of T_p in $\mathbb{R}^d$. Further, denote by T_p^2 the osculating space spanned by all first and second-order partial derivatives $\partial_i f, \partial_{ij} f$ at p , and let $N_p = T_p^2 \cap T_p^\perp$.

Since f is free, the dimensions of T_p , $T_p^\perp$, T_p^2 and N_p are n , $d - n$, $(n + \frac{n \cdot (n+1)}{2})$, and $\frac{n \cdot (n+1)}{2}$, respectively. We will write T , $T^\perp$, T^2 , and N for the corresponding bundles over $\mathbb{T}^n$.

Proof of 3.7. Recall that we have global periodic coordinates on $\mathbb{T}^n$.

Choose a metric tensor h on $\mathbb{T}^n$. Since all $\partial_i f$, $\partial_{ij} f$ are linearly independent, the equalities $\langle \partial_{ij} f, w \rangle = h_{ij}$ for all i and j uniquely define a N -vector field w . Alternatively, w can be defined as a linear combination of $\partial_i f$, $\partial_{ij} f$ such that

$$\textcircled{1} \quad \langle \partial_i f, w \rangle = 0, \quad -2 \cdot \langle \partial_{ij} f, w \rangle = h_{ij}$$

at each point.

Note that

$$\begin{aligned} L_{ij}(w) &= 2 \cdot Q_{ij}(f, w) = \\ &= \langle \partial_i f, \partial_j w \rangle + \langle \partial_j f, \partial_i w \rangle = \\ &= \partial_j \langle \partial_i f, w \rangle + \partial_i \langle \partial_j f, w \rangle - 2 \cdot \langle \partial_{ij} f, w \rangle. \end{aligned}$$

By $\textcircled{1}$, $M: h \mapsto w$ is the desired right inverse. $\square$

Note that the proof implies the following.

3.10. Observation. *The inverse M in Nash's lemma (3.7) can be chosen so that Mh is an N-field; that is, $Mh(p) \in N_p$ for any $p \in \mathbb{T}^n$.*

The following exercise provides a key to proof of Günther's lemma.

3.11. Exercise. *Let $f: \mathbb{T}^n \rightarrow \mathbb{R}^d$ be a free map. Given a tangent vector field v on $\mathbb{T}^n$ and a metric tensor h on $\mathbb{T}^n$, show that there is a unique vector N-field w such that*

$$L(v + w) = h.$$

In other words, given v , there is a right inverse $M_v: h \mapsto v + w$ of L with an N-field w .

E Proof of Günther's lemma

Many symmetric bilinear forms solve the equation $L\tilde{Q} = Q$. In particular, one can take $\tilde{Q} = MQ$, where M is provided by Nash's lemma (3.7). However, this choice of $\tilde{Q}$ does not satisfy the inequalities in Günther's lemma. These inequalities are important, they made it possible to use the contraction mapping theorem in the proof of perturbation lemma (3.9).

The actual construction of $\tilde{Q}$ relies on Exercise 3.11, which states that, given a tangent vector field v , we may choose M such that $w = Mh - v$ is an N-field. The choice of v provides additional freedom, which might seem unnecessary at first. Indeed, moving the map in a tangent direction does not change the Riemannian metric to first order; it only affects the parametrization.¹ However, as we will see, one can use v to hide the high-frequency terms (see ❷ and ❸ below), these terms are the central difficulty in Nash's original proof.

The Laplacian is defined by

$$\Delta u = \sum_s \partial_{ss} u,$$

where $\partial_1, \dots, \partial_n$ our standard choice of partial derivatives on $\mathbb{T}^n$.

Let $k \geq 2$ be an integer and let $v \in C^{k-2, \alpha}(\mathbb{T}^n, \mathbb{R})$. Fourier analysis implies the existence of a weak solution u to the equation $\Delta u - u = v$. By the Schauder estimate, it follows that $u \in C^{k, \alpha}(\mathbb{T}^n, \mathbb{R})$, and we obtain the following proposition.²

¹Compare to the DeTurck trick used for the Ricci flow.

²A proof of the Schauder estimates can be found in [5, Chapter 3]; some parts of it can be omitted in our case.

3.12. Proposition. Recall that $\alpha \in (0, 1)$ is a fixed number, say $\alpha = \frac{1}{2}$. Let $k \geq 2$ be an integer. Then

$$(\Delta - 1): C^{k, \alpha}(\mathbb{T}^n, \mathbb{R}) \rightarrow C^{k-2, \alpha}(\mathbb{T}^n, \mathbb{R}).$$

is a bi-Lipschitz isomorphism. In particular, the linear operator

$$(\Delta - 1): u \mapsto \Delta u - u$$

has a Lipschitz inverse

$$\textcircled{1} \quad (\Delta - 1)^{-1}: C^{k-2, \alpha}(\mathbb{T}^n, \mathbb{R}) \rightarrow C^{k, \alpha}(\mathbb{T}^n, \mathbb{R})$$

Note that $(\Delta - 1)^{-1}$ is a smoothing operator; it swallows a $C^{k-2, \alpha}$ -function and spits a $C^{k, \alpha}$ -function. Furthermore, $(\Delta - 1)^{-1}$ commutes with partial derivatives; that is,

$$(\Delta - 1)^{-1} \partial_i = \partial_i (\Delta - 1)^{-1}.$$

These two properties of $(\Delta - 1)^{-1}$ will play the key role in the following proof.

Proof of 3.8. Recall that $Q_{ij}(w, w) = \langle \partial_i w, \partial_j w \rangle$. Let us show that

$$\textcircled{2} \quad (1 - \Delta)Q_{ij}(w, w) = A_{ij}(w) + \partial_i A_j(w) + \partial_j A_i(w),$$

where each A_{ij} , A_i and A_j is a linear combination of the following quadratic forms

$$\textcircled{3} \quad w \mapsto \langle \partial_a w, \partial_b w \rangle, \quad w \mapsto \langle \partial_{ab} w, \partial_c w \rangle, \quad w \mapsto \langle \partial_{ab} w, \partial_{cd} w \rangle.$$

for all possible indexes a, b, c, d .

Indeed,

$$\begin{aligned} (1 - \Delta)Q_{ij}(w, w) &= \langle \partial_i w, \partial_j w \rangle - \\ &\quad - \sum_s [2 \cdot \langle \partial_{is} w, \partial_{js} w \rangle + \langle \partial_{iss} w, \partial_j w \rangle + \langle \partial_i w, \partial_{jss} w \rangle] = \\ &= \langle \partial_i w, \partial_j w \rangle - 2 \cdot \sum_s \langle \partial_{is} w, \partial_{js} w \rangle + 2 \cdot \langle \Delta w, \partial_{ij} w \rangle - \\ &\quad - \partial_i \langle \Delta w, \partial_j w \rangle - \partial_j \langle \Delta w, \partial_i w \rangle. \end{aligned}$$

Whence $\textcircled{2}$ follows.

Choose $A = A_i$ or $A = A_{ij}$ for some i and j . Note that

$$\begin{aligned} \textcircled{4} \quad \|A(w)\|_{0, \alpha} &\leq c_2 \cdot \|w\|_{2, \alpha}^2, \\ \|A(w)\|_{k-2, \alpha} &\leq c_2 \cdot \|w\|_{k, \alpha} \cdot \|w\|_{2, \alpha} + c_k \cdot \|w\|_{k-1, \alpha}^2. \end{aligned}$$

for some fixed constants $c_2, c_3, \dots$. Indeed, it is sufficient to check inequalities ④ for each form in ③, and the latter is straightforward.

Applying ①, we get

$$\begin{aligned} \textcircled{5} \quad & \|(1 - \Delta)^{-1} A(w)\|_{2,\alpha} \leq c_2 \cdot \|w\|_{2,\alpha}^2, \\ & \|(1 - \Delta)^{-1} A(w)\|_{k,\alpha} \leq c_2 \cdot \|w\|_{k,\alpha} \cdot \|w\|_{2,\alpha} + c_k \cdot \|w\|_{k-1,\alpha}^2; \end{aligned}$$

the constants $c_2, c_3, \dots$ might be different from those above, but each of them depends only on f .

Further, recall that

$$\textcircled{6} \quad L_{ij}(w) = -2 \cdot \langle \partial_{ij} f, w \rangle + \partial_i \langle \partial_j f, w \rangle + \partial_j \langle \partial_i f, w \rangle;$$

see Section 3D. Let us define $\tilde{Q}$ as a linear combination of $\partial_i f$ and $\partial_{ij} f$ for all $i \leq j$ such that

$$\begin{aligned} \textcircled{7} \quad & -2 \cdot \langle \partial_{ij} f, \tilde{Q}(w, w) \rangle = (\Delta - 1)^{-1} A_{ij}(w), \\ & \langle \partial_i f, \tilde{Q}(w, w) \rangle = (\Delta - 1)^{-1} A_i(w) \end{aligned}$$

for all i and j . Since f is free, $\tilde{Q}$ is well defined. Combining ⑥, ⑦, and ②, we get

$$L\tilde{Q} = Q.$$

Let $\{e_i, e_{ij}\}_{i \leq j}$ be the dual frame to $\{\partial_i f, \partial_{ij} f\}_{i \leq j}$ in osculating bundle T^2 . Note that

$$\tilde{Q}(w, w) = \sum_i (\Delta - 1)^{-1} A_i(w) \cdot e_i - \frac{1}{2} \cdot \sum_{i \leq j} A_{ij}(w) \cdot e_{ij}.$$

Since the maps $e_i, e_{ij}: \mathbb{T}^n \rightarrow \mathbb{R}^d$ are smooth, ⑤ and 3.6 imply the last inequality in the lemma for each integer $k \geq 3$. $\square$

Semisolutions

1.2. Choose a point $p \in \Omega$ and a chart at p .

Assume that the main theorem holds for d . Then components $g_{ij} = g(\partial_i, \partial_j)$ of any Riemannian metric can be written as

$$\textcircled{8} \quad \langle \partial_i w, \partial_j w \rangle = g_{ij},$$

for some smooth map $w: \Omega \rightarrow \mathbb{R}^d$.

We have $\binom{n+1}{2} = n \cdot (n+1)/2$ components g_{ij} ; each of them is a smooth function on Ω . The equation has d unknowns; these are the coordinate functions of w . This might convince you that d should be at least $\binom{n+1}{2}$. To prove this formally we will reduce the problem to an ordinary system of equations between jet-spaces of w and g at the point p ; the jet-space is the same as space of Taylor polynomials of a given degree at p .

Denote by $J^m(g)$ the Taylor polynomial of g at p of degree m . Note that this polynomial has $\binom{m+n}{n}$ coefficients from $n \cdot (n+1)/2$ -dimensional space. (Each coefficient is proportional to certain partial derivatives of g of order at most m at p .) Such polynomials form an open subset in a $\binom{n+1}{2} \cdot \binom{m+n}{n}$ -dimensional space.

Analogously, let $J^m(w)$ be the Taylor polynomial of w at p of degree m . Again, this polynomial has $\binom{m+n}{n}$ coefficients from d -dimensional space; therefore such polynomials form a space of dimension $d \cdot \binom{m+n}{n}$.

The equations $\textcircled{8}$ define a smooth map $J^{m+1}(w) \mapsto J^m(g)$. In other words, any partial derivatives of g of degree at most m can be expressed thru partial derivatives of w of degree at most $m+1$. Since the system of equations $\textcircled{8}$ have a solution for any metric g , there is $J^{m+1}(w)$ that maps to $J^m(g)$. Therefore, Sard's lemma implies that

$$d \cdot \binom{m+n+1}{n} \geq \binom{n+1}{2} \cdot \binom{m+n}{n}.$$

Since $\binom{m+n+1}{n} / \binom{m+n}{n} \rightarrow 1$ as $m \rightarrow \infty$, we get $d \geq \binom{n+1}{2}$.

Remark. The equation ❸ makes perfect sense for C^1 -smooth w , but in the proof we essentially used that w is C^m -smooth for sufficiently large m . Without this assumption, the statement does not hold.

Indeed Nash–Kuiper theorem [4, 6] says that ❸ has C^1 -smooth solution w if $d = n + 1$. Note that this is the optimal bound as if $d = n$, then the metric g is flat, which is not the case in general.

1.3. Consider square of a parallel 1-form α on $\mathbb{T}^2$ with kernel in irrational direction.

1.4. The only-if part is trivial.

Applying the length-preserving property to all arcs of curves in the coordinate directions, show that the equality

$$g_{ii} = \langle \partial_i w, \partial_i w \rangle$$

holds almost everywhere, and therefore everywhere.

Observe that

$$2 \cdot g_{ij} = g(\partial_i + \partial_j, \partial_i + \partial_j) - g(\partial_i, \partial_i) - g(\partial_j, \partial_j).$$

Using this formula with the argument above for curves in the direction ∂_i , ∂_j and $\partial_i + \partial_j$, show that the equality

$$g_{ij} = \langle \partial_i w, \partial_j w \rangle$$

holds almost everywhere, and therefore everywhere.

1.6+1.7. Spell the definitions.

1.13. Straightforward calculations.

2.5. By the definition of submanifold, we can cover Ω by a rectangular charts where Ω is written as the coordinate subspace of the first $k = \dim \Omega$ coordinates. Note that in each chart we can present the metric on Ω as a restriction of a Riemannian metric on the whole chart.

Extend this collection of charts to an atlas of $\mathbb{T}^n$ such that the images of the remaining charts do not intersect Ω . Equip these new charts with a constant Riemannian metric.

Choose a partition of unity subordinate to the covering by charts and use it to mix the metrics on the charts into a metric on $\mathbb{T}^n$.

2.6. Let $\iota: \Omega \rightarrow \mathbb{R}^d$ be an embedding provided by the Whitney theorem. Let us think that $\mathbb{R}^d$ is an affine subspace $x_{d+1} = 1$ in $\mathbb{R}^{d+1}$. The needed embedding can be found among maps $x \mapsto \varphi \cdot \iota$, where $\varphi: \Omega \rightarrow \mathbb{R}$ is a smooth positive function.

Try to write down the necessary conditions and convince yourself that there is such function φ .

2.7. Let $\mathbb{D} \subset \mathbb{R}^n$ be the unit n -disc and $\mathbb{D}^\circ$ is its interior. Construct a smooth map $r: \mathbb{R}^n \rightarrow \mathbb{S}^n$ such that $r|_{\mathbb{D}^\circ}$ is a smooth embedding and the set $\mathbb{R}^n \setminus \mathbb{D}^\circ$ is mapped to the south pole.

Since D_i is a smooth n -disc, there is a diffeomorphism $\iota_i: \mathbb{D} \rightarrow D_i$. The composition $r_i = r \circ \iota_i^{-1}$ satisfies all the first two conditions.

It remains to choose a metric h_i on $\mathbb{S}^n$ for each i that meets the last condition.

Choose sufficiently small metrics $\hat{h}_i$ on $\mathbb{S}^n$ so that

$$g > \hat{g} = \sum_i r_i^* \hat{h}_i.$$

Let $g_1 = g - \hat{g}$. Choose a partition of unity φ_i that is subordant to the covering $\{D_i\}$; that is, $\sum \varphi_i = 1$ and the support of φ_i lies in D_i for each i . Observe that $h_i = \hat{h}_i + \iota_i^*(\varphi_i \cdot g_1)$ meets the last condition.

3.1. Show and use that a smooth map $f: \Omega \rightarrow \mathbb{R}^d$ is free at $p \in \Omega$ if there is no affine subspace A of dimension less than $\frac{n \cdot (n+3)}{2}$ that has the second order contact with f at p ; that is,

$$\rho \circ f(x) = o(|p - x|^2),$$

where $\rho(z)$ denotes the distance from z to A .

Alternatively, express the partial derivatives in one chart thru the partial derivatives in the other chart and make a conclusion.

3.2. The definition of free map requires linear independence of certain partial derivatives. Count them.

3.3. Spell the definition.

3.4. This can be done by calculating the partial derivatives. Alternatively, one may apply the observation with contact order with an affine subspace from 3.1.

In the higher dimensional case, one may take

$$\bigoplus_{i \leq j} \Theta_{x_i + x_j},$$

where x_i are coordinate functions. The same proof should work.

3.5. Use the definition of free map and 3.1.

3.6. Apply the product rule for derivatives.

3.11. Note that the equation can be written as $L(w) = \hat{h}$, where $\hat{h} = h - L(v)$. It remains to apply the Nash's lemma.

Index

$(d\psi)^2$, 8

$\times w$, 5

dw , 5

$w * g$, 5

differential, 5

free map, 17

induced metric tensor, 6

isometric map, 6

length-preserving, 6

metric tensor, 5

Nash's reduction, 15

Nash's twist, 8

pullback, 6

regular map, 17

Riemannian manifold, 6

Riemannian metric tensor, 6

Schauder estimate, 22

smooth n -disc, 7

star, 8

support, 7

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